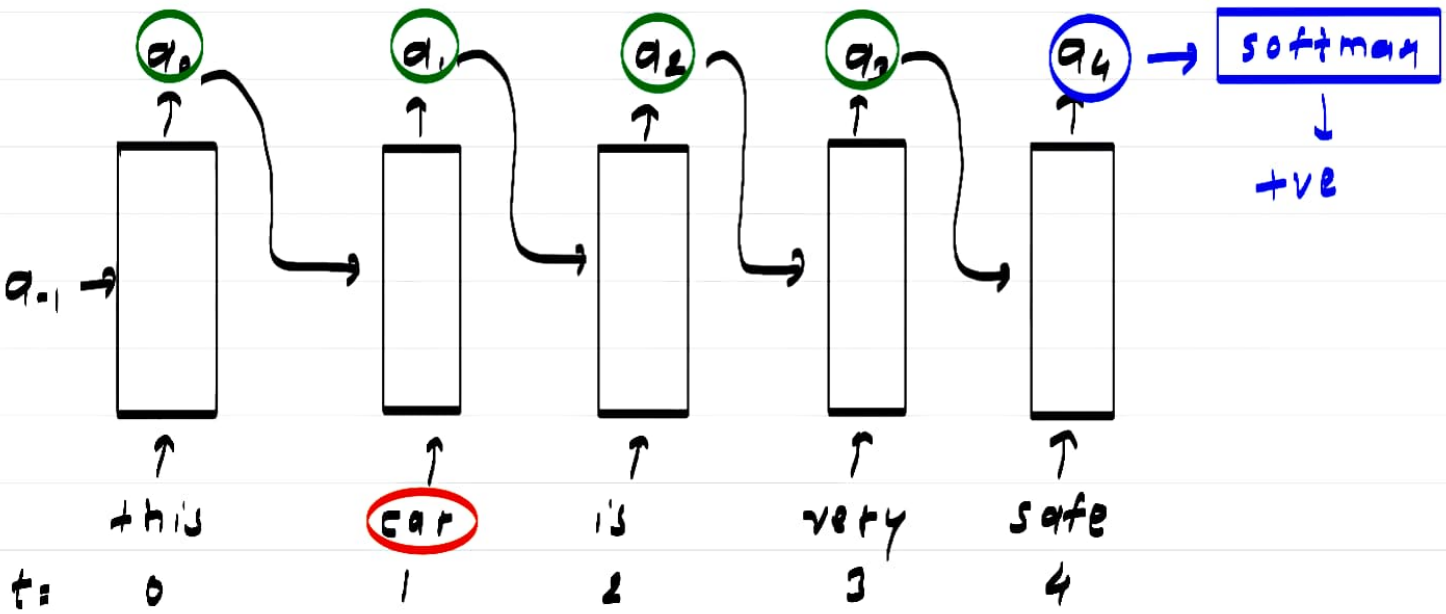
NLP

18-06-26

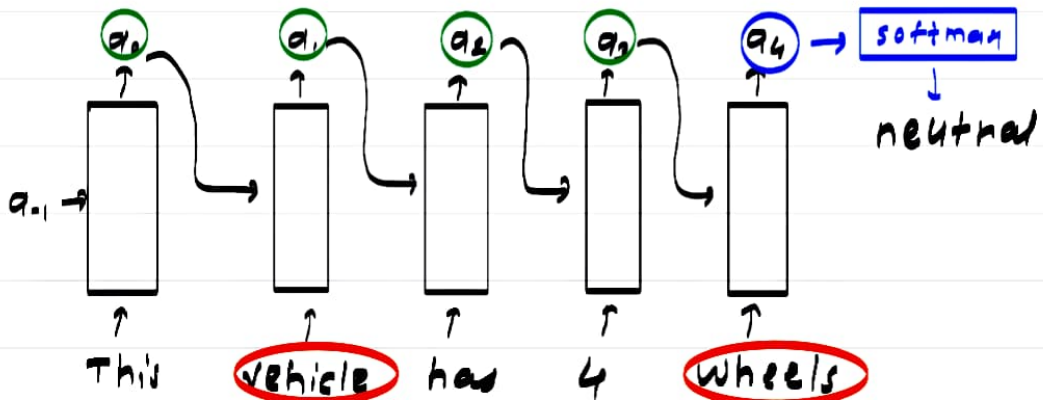
Word Embeddings:

- its the vector representation of the word / token which is generated at an intermediate step in an NLP model.

eg. text classification



the output vectors a_0 a_1 a_2 a_3 a_4 have captured the meaning of each token in context of given sequence. a_0 a_1 a_2 a_3 a_4 are word embeddings.



- When NLP models are trained over large amount of text data, they start to understand the meaning of token as well as how some tokens/entities are related to each other.

- Using training data these models also become aware of real world entities.

eg. in example above.

car and vehicle will have their vector representations very similar to each other

we measure this similarity using cosine similarity

car $\rightarrow V_1$ vehicle $\rightarrow V_2$

V_1 V_2 are vector representations of these words generated by NLP model.

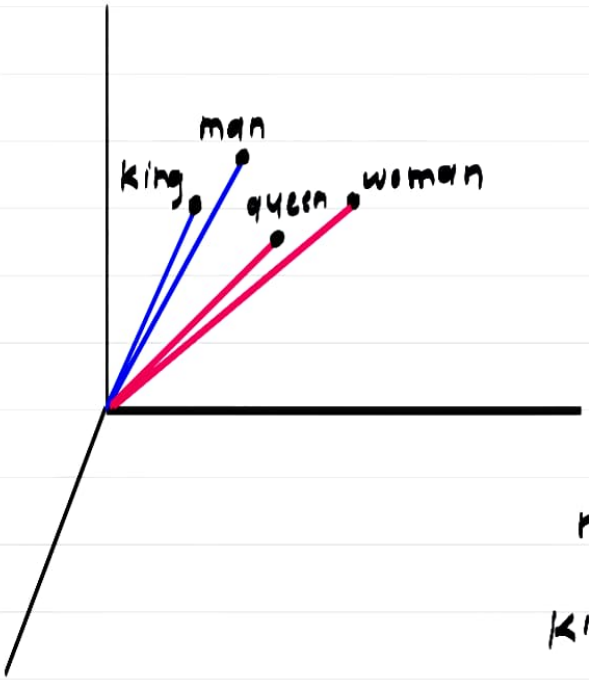
$$\text{cosine sim} = \frac{V_1 \cdot V_2}{|V_1|\,|V_2|}$$

$$-1 \leq \cos \text{ sim} \leq 1$$

cos sim ≈ 0 $\longrightarrow$ V_1 V_2 are not similar
cos sim ≈ 1 $\longrightarrow$ V_1 V_2 are highly similar
cos sim ≈ -1 $\longrightarrow$ V_1 V_2 are opposites of each other

word embeddings - vector representation of tokens
word embeddings are byproduct of NLP task.

vectors in 3 dim space



king : $V_1 = [V_{01}\ V_{11}\ V_{21}]$

man : $V_2 = [V_{02}\ V_{12}\ V_{22}]$

queen : $V_3 = [V_{03}\ V_{13}\ V_{23}]$

woman : $V_4 = [V_{04}\ V_{14}\ V_{24}]$

man - woman $\approx$ king - queen

king - man + woman $\approx$ queen

dot product of 2 vectors $V_1\ V_2$

$V_1 \cdot V_2 = |V_1|\,|V_2| \cos(\theta) = V_{01} \times V_{02} + V_{11} \times V_{12} + V_{21} \times V_{22}$ ✓

$$\cos(\theta) = \frac{V_1 \cdot V_2}{|V_1|\,|V_2|}$$

where θ is angle betn the vectors

$\cos(\theta)$: cosine similarity

smaller θ ⟶ higher similarity

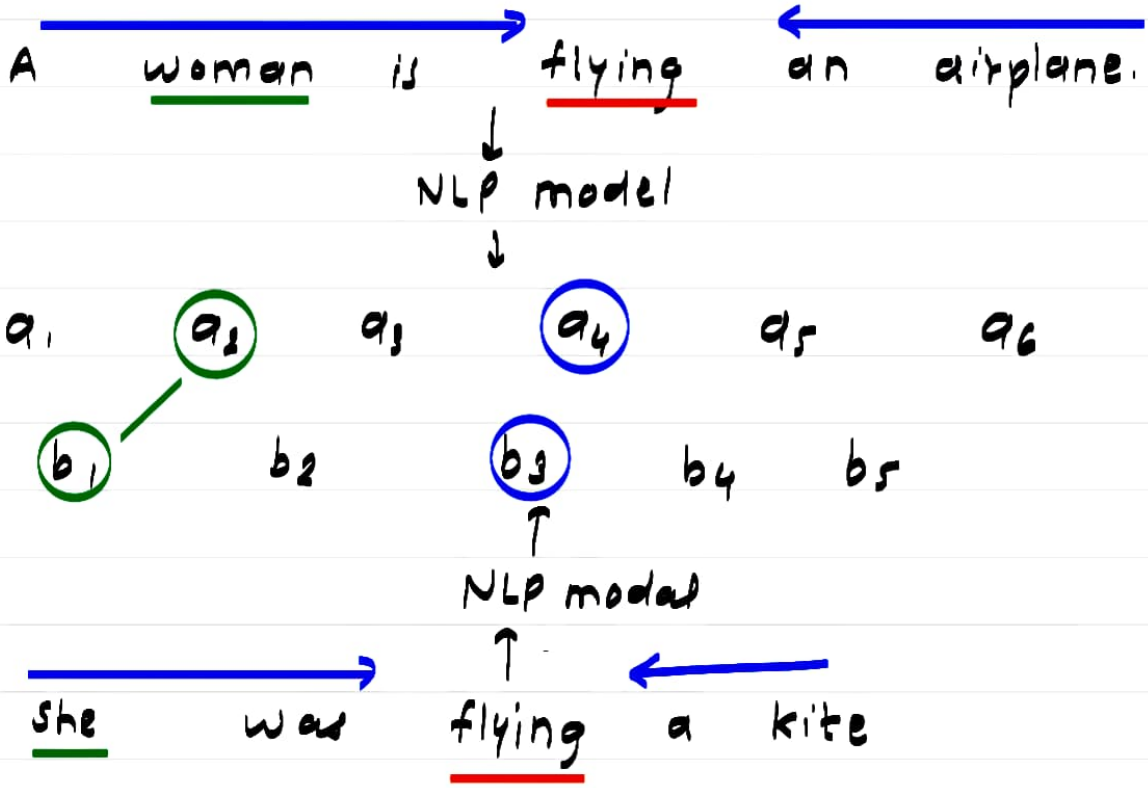
These learned word embeddings are very useful for transfer learning, retrieval, etc.

↓

RAG

↓

Retrieval Augmented Generation



a_2 and b_1 will have high cosine similarity as they have similar meaning

a_4 and b_3 - though they represent the exact same word, there will be minor differences in them due to different context.

Sentence / text embedding:

in above example - individual vectors are representing one token each

average of all word embeddings gives vector representation of complete sentence

sentence embedding = average of all word embeddings in given sentence

If two sentences have similar meaning, there sentence embeddings will be also similar.

eg.

S1- Trees are important for enviroment

↓ NLP model

a_1 a_2 a_3 a_4 a_5

sent embed : $a = avg(a_1, a_2, a_3, a_4, a_5)$

S2- Environment is highly dependent on trees

↓ NLP model

b_1 b_2 b_3 b_4 b_5 b_6

sent embed: $b = avg(b_1, b_2, b_3, b_4, b_5, b_6)$

As the meaning of S1 and S2 is highly similar

cosine similarity betⁿ a and b will be high

look text articles

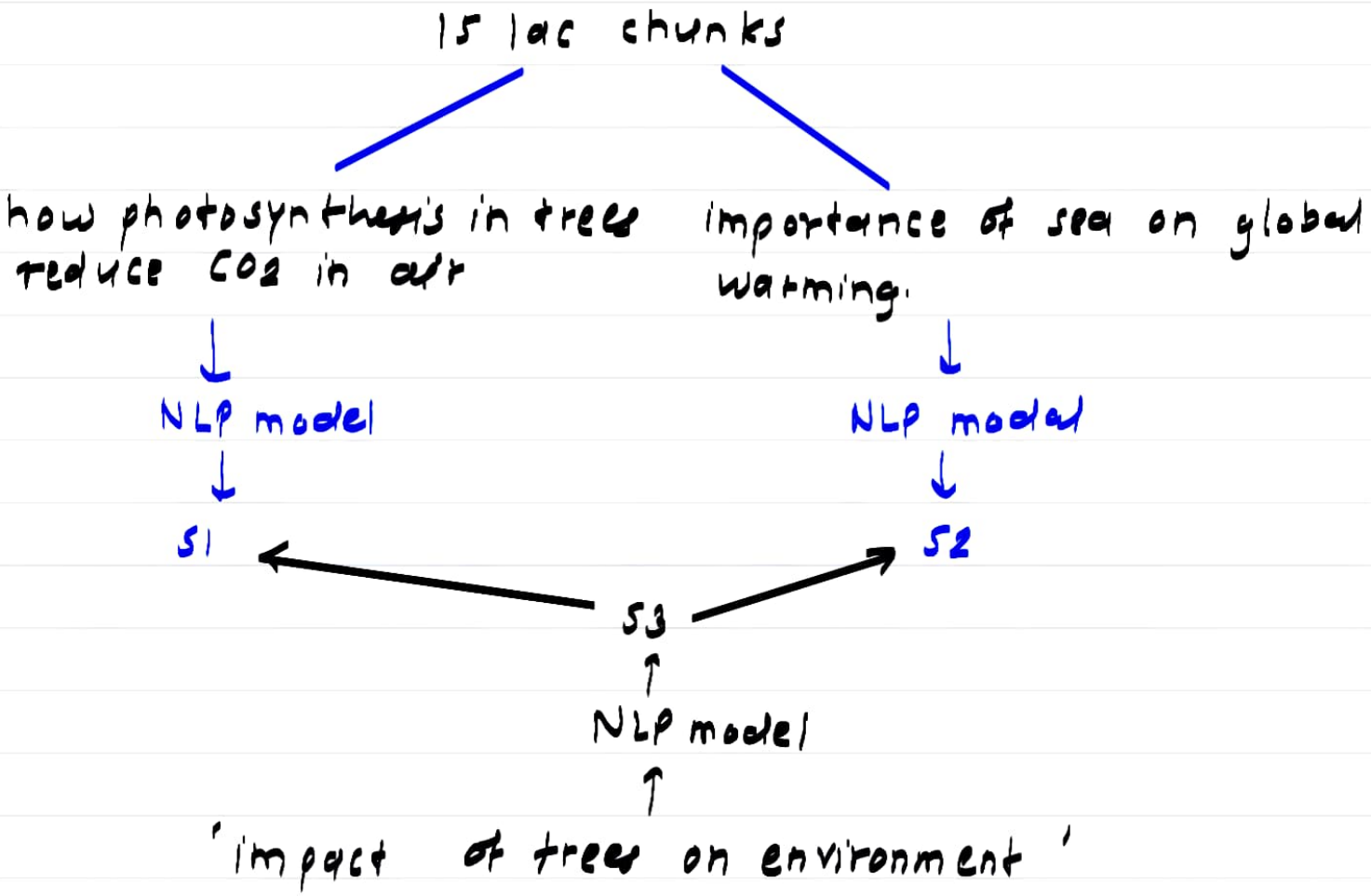
'impact of trees on environment'

↓ chunk articles into paragraphs

1 lac articles → 15 lac paragraph chunks

↓ NLP model

15 lac embeddings



cosine sim S1 and S3 > cosine sim S2 and S3

0.6 > 0.4